YEAR 9 MATHS · CHAPTER 6 · MIND MAP

Indices and surds

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1 Index notation

2^5: 2 is the base, 5 is the index (how many 2s multiply).

$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$

Prime factors: $72 = 2^3 \times 3^2$

2 Multiplying and dividing

Same base: multiply, add the indices. Divide, subtract them.

$a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$

$x^3 \times x^4 = x^7$ $6a^5 \div 2a^2 = 3a^3$

Watch out: $2^3 \times 2^4 = 2^7$, not 4^7

3 Power of a power, zero index

Power of a power: multiply the indices. Anything to the power 0 is 1.

$(a^m)^n = a^{mn}$: $(x^2)^5 = x^{10}$

$7^0 = 1$ $5x^0 = 5$

4 Brackets and fractions

The power applies to everything inside the brackets.

$(2x^3)^2 = 4x^6$

$\left(\frac{x}{3}\right)^2 = \frac{x^2}{9}$

Watch out: square the 2 too: $(2x)^2 = 4x^2$

5 Negative indices

A negative index means one over: $a^{-n} = \frac{1}{a^n}$.

$2^{-3} = \frac{1}{8}$ $x^{-2} = \frac{1}{x^2}$ $\frac{1}{a^{-2}} = a^2$

6 Scientific notation

Write numbers as $a \times 10^n$ with $1 \leq a < 10$.

$45\,000 = 4.5 \times 10^4$

$0.0062 = 6.2 \times 10^{-3}$

Watch out: big numbers get a positive power, small ones a negative power.

7 Scientific notation and significant figures

Round the first number to the significant figures asked for.

$0.000\,456\,7$ **to 2 s.f.:** 4.6×10^{-4}

$3\,870\,000$ **to 2 s.f.:** 3.9×10^6

$(2.4 \times 10^5) \times (3 \times 10^3) = 7.2 \times 10^8$

8 Fractional indices and surds

$a^{\frac{1}{2}} = \sqrt{a}$, $a^{\frac{1}{3}} = \sqrt[3]{a}$. A surd is a root that isn't a whole number, like $\sqrt{2}$.

$9^{\frac{1}{2}} = 3$ $8^{\frac{1}{3}} = 2$

Simplify: $\sqrt{12} = \sqrt{4} \times \sqrt{3} = 2\sqrt{3}$

9 Simple operations with surds

Add or subtract like surds only. Multiply: multiply the numbers outside and inside.

$3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$ $\sqrt{3} \times \sqrt{5} = \sqrt{15}$

$2\sqrt{3} \times 4\sqrt{2} = 8\sqrt{6}$

$\sqrt{8} + \sqrt{18} = 2\sqrt{2} + 3\sqrt{2} = 5\sqrt{2}$

Watch out: $\sqrt{2} + \sqrt{3} \neq \sqrt{5}$